Thirteen Variables from Direct Elliptic Index Comparison

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Abstract

Under the weak rational *abc* inequality with exponent 121/100 and the stated elliptic, pole-realization, and prime-pattern inputs, we construct a computable family of integer polynomials in thirteen variables, with degree at most 11774654098433. The elliptic comparison recovers the input itself as an integer index, while a separate positive quadratic expression controls its height. This permits a sixth-power comparison scale. The same construction with a fourth-power scale has degree at most 11212784195585 under the weak inequality with exponent 16/15. Ordinary Szpiro implies the first hypothesis; ordinary rational *abc* implies both. Every sufficiently large fixed odd coefficient shift works, but the sufficient thresholds are ineffective. The input zero is included by a single polynomial factor, without an additional variable. Complete finite coefficient specifications, independent exact conversion checks, and uniform degree and support bounds accompany the proofs.

1 The families and their hypotheses

All variable counts include the input. For a reduced rational number u/v, with $v > 0$, write

$$h(u/v) = \log \max(|u|, v).$$

Fix one of the two pairs

$$(\kappa, d) = (121/100, 6) \quad \text{or} \quad (16/15, 4).$$

The additional height hypothesis is that a constant $K_\kappa \geq 1$ exists such that

$$c \leq K_\kappa \operatorname{rad}(abc)^\kappa \qquad (a + b = c,\ a, b, c > 0 \text{ pairwise coprime}). \tag{1}$$

Moving the negative terms to the other side gives the corresponding statement for a coprime nonzero signed triple of sum zero, with the largest absolute value on the left.

Set

$$K = \mathbb{Q}(\sqrt{2}), \quad D = (3 - \sqrt{2})^2, \quad E : y^2 = x^3 - Dx,$$
$$P = \left(\frac{57}{8} - \frac{7}{2}\sqrt{2}, -\frac{165}{8} + \frac{417}{32}\sqrt{2}\right).$$

In addition to (1), retain the following arithmetic inputs with exactly their stated scope in [3, 2].

1. $2E(K) = \mathbb{Z}P$, with the indicated non-torsion generator and the source index-height bound. The direct point chart below therefore describes the nonzero multiples of P.

2. The pole system, for $b \neq 0$ and $B_0 = 17b$,

$$p^2 = B_0(1 - e)(1 + 3e), \qquad B_0 q^2 = (s^2 - B_0 e(B_0 - b^2))(s^2 + 3B_0 e(B_0 - b^2)),$$

excludes negative odd valuations of b outside $\{2,3,17\}$. For a prepared input

$$0<b<17,\quad b \text{ has square reduced denominator},\quad b\in\mathbb{Q}_2^{\times 2}\cap\mathbb{Q}_3^{\times 2}\cap\mathbb{Q}_{17}^{\times 2},\\ v_2(b)\geq 3,\quad v_3(b)\geq 1,\quad v_{17}(b)\geq 2, \tag{2}$$

the retained restricted-converse and descent hypotheses supply a witness with $pqs\neq 0$. The nonzero-witness consequence and its single conic chart are proved in [2, Lemma 2].

3. The simultaneous prime-pattern input supplies the specified prime contact numerators, Gaussian congruences, conjugate reduction orders, fixed divisibility and avoidance, and compatible real approximation for a fixed finite list of rational contact coefficients. Its precise quantified version and the rational-contact and fixed-avoidance consequences of Sun's theorem are given in [3]. The even number of contacts below permits arbitrary positive rational scales: Section 6 checks the square factors directly. The order of choices is respected there.

No modular-realization or polynomial contact-height premise is used. Lemma 2.1 supplies the contact-height estimate from (1).

The proved one-point case of the aggregate trace-height theorem [5] gives constants $A>0$, $J\geq 0$, depending only on E,P, such that

$$n^2\leq Ah(T(n))+J,\qquad T(n)=\operatorname{Tr}_{K/\mathbb{Q}}x([n]P)\quad (n\in\mathbb{Z}\setminus\{0\}). \tag{3}$$

This applies von Buhren's theorem after checking the absence of constant abelian translates for the fixed trace function. Its additive constant is ineffective. It is a proved height theorem, not an additional conjecture. For a nonzero multiple of P, the abscissa is a positive square in both real embeddings; the curve equation then gives $x>3\mp\sqrt{2}$ in the respective embeddings, hence $T(n)>6$.

Theorem 1.1. *Retain the arithmetic inputs above. For each of the two pairs (κ,d), there is a computable family*

$$F_B^{(d)}\in\mathbb{Z}[X,Y_1,\dots,Y_{12}],\qquad B\geq 3 \text{ odd},$$

with the following properties.

1. *Each member has a complete finite integer coefficient specification. Uniformly in B, its degree and number of ordinary monomials satisfy*

d	*degree upper bound*	*support upper bound*
6	11774654098433	$<2^{508}$
4	11212784195585	$<2^{507}$.

   In particular, both degree bounds are less than 2^{44}.

2. *Under (1) for the indicated κ, there is a constant $B_*^{(d)}$ such that every odd $B\geq B_*^{(d)}$ has rational input projection exactly $\mathbb{Z}$.*

3. *No numerical sufficient value of $B_*^{(d)}$ is determined. The supplied $B=3$ members are coefficient-verification fixtures, not certified defining members.*

The fixed coefficient B is never a quantified variable.

Ordinary rational *abc* implies both displayed weak inequalities. Oesterlé proves that ordinary discriminant–conductor Szpiro implies weak *abc* with every exponent $6/5+\epsilon$ [1, Section I.3, p. 169]. Taking $\epsilon=1/100$ therefore supplies the sixth-power family, even from the semistable case of ordinary Szpiro. No implication from ordinary Szpiro to the stronger exponent 16/15 is asserted. None of these implications removes the retained elliptic and pole-realization premises.

2 Height estimates

Put

$$\Theta = \left\{0, -\frac{1}{2}, -\frac{1}{4}, -\frac{2}{5}, -1, -\frac{1}{3}\right\}, \qquad S_0 = \{p : p \le 76176\}.$$

For $t \in \mathbb{Q} \setminus \Theta$, let $M_0(t)$ be the product of primes outside S_0 for which $v_p(t-\theta)$ is positive and odd for some $\theta \in \Theta$. Define

$$\gamma = 8/\kappa - 6, \qquad \alpha = d/\kappa - 3.$$

Both numbers are positive for either pair used here.

Lemma 2.1. *Under (1), there is a constant $C_0 \ge 0$, independent of every later coefficient shift, such that*

$$h(t) \le \gamma^{-1} \log M_0(t) + C_0 \qquad (t \in \mathbb{Q} \setminus \Theta).$$

Proof. Consider the degree-four rational function

$$\varphi(z) = \frac{z(z+2)^3}{(2z+1)^3}, \qquad \varphi(z) - 1 = \frac{(z-1)^3(z+1)}{(2z+1)^3}.$$

Its complete inverse image of $\{0, 1, \infty\}$ is $\{0, -2, -1/2, 1, -1, \infty\}$. For a primitive integer pair (x, y) away from these six points, put

$$A = x(x+2y)^3, \quad C = y(2x+y)^3, \quad A - C = (x-y)^3(x+y),$$

$$G(x,y) = xy(x+2y)(2x+y)(x-y)(x+y).$$

The polynomial identity for $A - C$ follows by expansion. A prime dividing both A and C must divide both $x + 2y$ and $2x + y$, so it is 3. At that prime x, y are units and the smaller of those two valuations is one, because $2(x + 2y) - (2x + y) = 3y$. Hence $\gcd(A, C)$ is either 1 or 27. After division by this gcd, the three integers $A, C, A - C$ are pairwise coprime, and their radical is supported on G.

Write $H = \max(|x|, |y|)$. One has $\max(|x + 2y|, |2x + y|) \ge H$. If the corresponding prefactor $|x|$ or $|y|$ is at least $H/4$, this gives $\max(|A|, |C|) \ge H^4/4$. Otherwise the other prefactor is H and its linear factor is at least $H/2$, giving $\max(|A|, |C|) \ge H^4/8$. Consequently

$$h(\varphi(x/y)) \ge 4 \log H - \log 216.$$

The weak inequality therefore implies

$$\log \operatorname{rad}(G(x,y)) \ge \frac{4}{\kappa} \log H - \frac{\log 216 + \log K_\kappa}{\kappa}.$$

For a nonzero integer N, let $\operatorname{sf}(N)$ be the product of primes of odd valuation in N. Prime by prime, $\operatorname{rad}(N)^2 \le |N| \operatorname{sf}(N)$. Also $|G(x,y)| \le 36H^6$. Outside the fixed set S_0, two of its six primitive linear factors cannot vanish together. Thus its squarefree part outside S_0 is precisely the odd contact radical for the six displayed points. The contribution from S_0 is bounded by the fixed product of its primes. Combining these estimates gives

$$\log M(z) \ge \left(\frac{8}{\kappa} - 6\right) h(z) - C = \gamma h(z) - C.$$

Finally $z = 2 + 1/t$ sends the six contacts in Θ to the six points above. Its integral matrix has determinant -1; its inverse is integral too, and both matrices have maximum row sum three. Hence it changes logarithmic height by at most $\log 3$. Substitution of a primitive numerator-denominator pair shows that the odd contact radicals agree outside S_0. This proves the lemma after adjusting the fixed constant. □

For $u, v \in \mathbb{Q}$, let

$$h_2(u,v) = h(1:u:v) = \log \max(D_0, |A'|, |B'|),$$

where $D_0 > 0$, $u = A'/D_0$, $v = B'/D_0$, and $\gcd(D_0, A', B') = 1$. Each individual height is at most h_2.

Lemma 2.2. *For an even d with $\alpha = d/\kappa - 3 > 0$, one has*

$$h_2(u,v) \le \alpha^{-1} h(u^d + v^d) + C_1 \qquad (u, v \in \mathbb{Q}),$$

where one may take

$$C_1 = \frac{\alpha+1}{d\alpha}\log 2 + \frac{1}{\kappa\alpha}\log K_\kappa.$$

Proof. If one coordinate is zero, $h_2(u,v) = h(u^d+v^d)/d$; if both are zero all heights vanish. Otherwise write $g = \gcd(|A'|, |B'|)$, $A_0 = A'/g$, and $B_0 = B'/g$. Then $\gcd(A_0, B_0) = \gcd(g, D_0) = 1$. Put

$$H = \max(|A_0|, |B_0|), \qquad N = A_0^d + B_0^d, \qquad H^d \le N \le 2H^d.$$

Set $k = \gcd(N, D_0^d)$, $m_0 = N/k$, and $n_0 = D_0^d/k$. The reduced fraction for $u^d + v^d$ is m/n_0, where $m = g^d m_0$. Let $M = \max(m, n_0)$, so its height is $\log M$. Every prime dividing N either divides D_0 or survives in m_0, hence $\mathrm{rad}(N) \le D_0 m_0$. Applying (1) to the coprime positive triple A_0^d, B_0^d, N gives

$$H^{d/\kappa - 2} \le K_\kappa^{1/\kappa} D_0 m_0.$$

Since $D_0^d = n_0 N/m_0 \le 2n_0 H^d/m_0$, this yields

$$H^\alpha \le 2^{1/d} K_\kappa^{1/\kappa} m_0^{(d-1)/d} n_0^{1/d}.$$

Also $g = (m/m_0)^{1/d}$, so

$$\max(gH, D_0) \le 2^{1/d} H (M/m_0)^{1/d}.$$

Combining the logarithms proves

$$\begin{aligned} h_2(u,v) &\le C_1 + \frac{d-1-\alpha}{d\alpha}\log m_0 + \frac{1}{d\alpha}\log n_0 + \frac{1}{d}\log M \\ &\le C_1 + \alpha^{-1}\log M. \end{aligned}$$

The three coefficients are nonnegative and sum to $1/\alpha$; indeed $d - 1 - \alpha = d(1 - 1/\kappa) + 2 > 0$. This calculation includes the common numerator factor and all cancellation with the denominator. □

Lemma 2.3. *For every rational X and integer $B \ge 3$, put*

$$\beta = X^2 + B. \tag{4}$$

Then

$$\beta \ge B, \qquad 2h(X) \le h(\beta).$$

Proof. Write $X = u/v$ in lowest terms with $v > 0$. Since $\gcd(u^2 + Bv^2, v^2) = 1$, the displayed numerator and denominator of $\beta = (u^2 + Bv^2)/v^2$ are already reduced. Moreover

$$u^2 + Bv^2 \ge \max(u^2, v^2).$$

Since $\beta \ge B > 1$, its height is the logarithm of this positive numerator. This proves the height inequality, including $u = 0$. There is no cancellation factor. □

The integer label compared with the elliptic index is X. The separate quantity β appears only in the height scale; there is no need to infer integrality of X from integrality of β.

3 The complete rational source

Fix an odd integer $B \geq 3$ and use (4). Retain three pairs of rational coordinates, one for each of the points A, H, U. For any such pair (h_0, h_1) put

$$h = h_0 + h_1\sqrt{2}, \quad x = h^2, \quad j^2 = h^4 - D, \quad y = hj.$$

The direct point-chart theorem of [2] proves that these are exactly the nonzero multiples of P. Each finite choice j is encoded by one scalar monic quartic as follows. Write $h^4 - D = a + b\sqrt{2}$, put $c = 2a + b$, $d = -a - b$, and write $(c + d\sqrt{2})^2(a + b\sqrt{2}) = a' + b'\sqrt{2}$. Then

$$Y^4 - 4a'Y^2 + 8(b')^2 = 0, \quad j = \frac{Y/2 + (b'/Y)\sqrt{2}}{c + d\sqrt{2}}.$$

Its discriminant is $2^{15}(b')^2(2a^2 - b^2)^4(a^2 - 2b^2)^2$, nonzero at every rational pair (h_0, h_1). Indeed $h^4 - D \neq 0$ by its norm, the two quadratic norm factors are anisotropic over $\mathbb{Q}$, and $b' = -4a^3 + 6ab^2 + 3b^3$ is anisotropic by Eisenstein at 3. The square-root chart is complete because its transformed rational component is $2(u^3 - 2v^3)$ for $j = u + v\sqrt{2}$, which is nonzero for $j \neq 0$. These are algebraic facts independent of the height hypothesis.

Compute $V_0 = H - P$ and $V_1 = U - A$ by the usual affine addition formulas. Put $f = 1/x$, $z = -x/y$, let σ be conjugation of $K/\mathbb{Q}$, and define

$$L = \frac{z(U)}{z(H)}, \qquad L' = \sigma\left(\frac{z(V_1)}{z(V_0)}\right).$$

Use precisely the six field-valued tests

$$D_1 = f(H),\ D_2 = f(U),\ D_3 = \sigma f(V_0),\ D_4 = \sigma f(V_1),\ D_5 = L - L',\ D_6 = L - X. \tag{5}$$

Every test is assigned order one. No scalar-height pole or Lucas relation is present.

Let $T = \operatorname{Tr} x(A)$, and use the two positive rational scales

$$a_0 = 1, \qquad a_1 = \beta^d + T^d.$$

Retain one rational slope λ. For $r = 0, 1$, set

$$t_r = a_r\lambda, \qquad \delta_r = t_r - 1, \qquad F_r = \prod_{\theta\in\Theta}(t_r - \theta), \qquad R_r = \frac{F_r}{\delta_r^8}, \tag{6}$$

$$W_r = \prod_{j=1}^{6}\left(1 + \frac{D_j^2}{\delta_r^4 R_r}\right), \qquad b_r = \frac{R_r}{\operatorname{Tr}(W_r^{-1})^2}. \tag{7}$$

All displayed divisions, affine-chart denominators, F_r, δ_r, $\operatorname{Norm}(W_r)$, and $\operatorname{Tr}(W_r^{-1})$ are guarded nonzero. These conditions are part of the source.

For each $b = b_r$, retain independent rational parameters τ_r, s_r. Put $B_0 = 17b$ and

$$e = \frac{\tau_r^2 - B_0}{\tau_r^2 + 3B_0}, \qquad q^2 = \frac{(s_r^2 - B_0e(B_0 - b^2))(s_r^2 + 3B_0e(B_0 - b^2))}{B_0}.$$

Guard the denominator and require $q \neq 0$. This reconstructs the first pole ordinate as $p = -4B_0\tau_r/(\tau_r^2 + 3B_0)$. Each rational-square equation $q^2 = N_0/D_0$ is made monic by $Z = D_0q$, giving $Z^2 = N_0D_0$ with $Z \neq 0$ and $D_0 \neq 0$. No new unrestricted rational parameter is used for these roots.

4 The local implication

Lemma 4.1. *For any rational solution of the guarded source, write $A = [n]P$, $H = [h]P$, $U = [u]P$. If $X \neq n$ and N is the positive absolute numerator of $X - n$ in lowest terms, then*

$$M_0(t_r) \mid N \qquad (r = 0, 1).$$

Proof. Fix a prime $p \mid M_0(t_r)$ and abbreviate $R = R_r$, $W = W_r$. Outside S_0 only one contact factor has positive valuation, δ_r is a unit, and $v_p(R)$ is odd because its denominator is an eighth power. For each place $\mathfrak{p}$ above p, the two terms of a factor of W have valuations of different parity. Hence

$$v_{\mathfrak{p}}(W) = \sum_j \min\{0, 2v_{\mathfrak{p}}(D_j) - v_p(R)\} \leq 0.$$

Put $w = \max_{\mathfrak{p}|p} v_{\mathfrak{p}}(W) \leq 0$. Then $v_p \operatorname{Tr}(W^{-1}) \geq -w$. The valuation of $b = R/\operatorname{Tr}(W^{-1})^2$ is odd, so the pole obstruction gives

$$0 < v_p(b) \leq v_p(R) + 2w.$$

Thus $\nu = v_p(R) > 0$ and $w > -\nu/2$. At a place attaining w, each summand in the valuation of W is at least w. It follows that $v_{\mathfrak{p}}(D_j) > \nu/4 > 0$, so all six tests have valuation at least one at that place.

The first four tests put H, U into the formal group at $\mathfrak{p}$ and V_0, V_1 into it at the conjugate place. At these good unramified places the formal logarithm satisfies

$$\log_E(Q) = z(Q)(1 + O(p^4)).$$

Indeed its first omitted term is a fixed multiple of $Df(Q)^2$; $v_p(f(Q)) \geq 2$ in the formal group. The remaining coefficient estimates, uniformly for $p > 76176$, are those proved in the formal-logarithm paper [6]. The exact logarithmic ratios are u/h and $(u - n)/(h - 1)$, respectively.

The affine guards imply $h \neq 0, 1$. At least one of $h, h-1$ is a p-adic unit, so the corresponding exact ratio is integral. Relative precision four makes its truncated ratio integral. The test $L - L' \equiv 0 \pmod{p}$ makes the other truncated ratio integral as well; the same relative precision then makes the other exact ratio integral. Relative errors are now absolute errors of valuation at least four. Consequently

$$L \equiv u/h, \qquad L' \equiv (u - n)/(h - 1), \qquad L \equiv L' \pmod{p}.$$

Multiplying by the integral numbers $h, h - 1$ and subtracting yields $L \equiv n \pmod{p}$. The target test gives $L \equiv X \pmod{p}$. In particular X is p-integral, and p divides the reduced numerator of $X - n$. This holds at every consumed prime and proves the divisibility. The same argument includes inert primes; if their formal-group requirements are incompatible, no source solution occurs at that prime. □

5 Soundness and a uniform sufficient shift

Suppose a rational source solution has $X \neq n$, and let N be the positive absolute numerator of $X - n$. The point chart gives $n \neq 0$, because A is a nonzero multiple of the non-torsion point P. Lemmas 4.1 and 2.1 give

$$h(\lambda), h(a_1\lambda) \leq \gamma^{-1} \log N + C_0.$$

The nonzero slope and the quotient-height inequality therefore imply

$$h(a_1) \le 2\gamma^{-1} \log N + 2C_0.$$

Apply Lemma 2.2 to $\beta, T(n)$ and set

$$\eta = h_2(\beta, T(n)), \qquad q = \frac{2}{\gamma\alpha}, \qquad C = \frac{2C_0}{\alpha} + C_1.$$

Then

$$\eta \le q \log N + C. \tag{8}$$

The constants depend on the fixed contact function, d, and the weak-*abc* constant, but not on B, X, or any witness. The relevant rational numbers are

(κ, d)	γ	α	q	$q/2$
$(121/100, 6)$	$74/121$	$237/121$	$14641/8769$	$14641/17538$
$(16/15, 4)$	$3/2$	$3/4$	$16/9$	$8/9$.

In both cases $q/2 < 1$.

Writing $X = u/v$ in lowest terms gives

$$N = |u - nv| \le e^{h(X)}(1 + |n|), \qquad h(X) \le \frac{1}{2}h(\beta) \le \frac{1}{2}\eta.$$

Consequently (8) implies

$$\eta \le c \log(1 + |n|) + C', \qquad c = \frac{q}{1 - q/2}, \quad C' = \frac{C}{1 - q/2}.$$

By the one-point trace-height estimate (3),

$$|n|^2 \le Ac \log(1 + |n|) + AC' + J \le Ac|n| + AC' + J.$$

Choose an integer $M > Ac + \sqrt{AC' + J}$. It follows that $|n| < M$, and hence

$$h(\beta) \le \eta \le c \log(1 + M) + C' =: H_*.$$

If $B_* > e^{H_*}$, every odd $B \ge B_*$ contradicts

$$h(\beta) \ge \log \beta \ge \log B > H_*.$$

Thus every rational source solution for such a shift has $X = n \in \mathbb{Z} \setminus \{0\}$. The proof is uniform in the shift; it supplies no effective numerical value of B_*.

6 Completeness for every nonzero integer

Fix $X = n \in \mathbb{Z} \setminus \{0\}$ and an odd $B \ge 3$. Fix $A = [n]P$ and the two scales before choosing any prime. The scales are positive and distinct, since $\beta \ge 3$ and d is even. Temporarily put $H = [h]P$ and $U = [hn]P$. The sign and parity of n are unrestricted.

Choose a nonexceptional $h_* \in 64\mathbb{Z}$, and work in the real orbit closure of $64\mathbb{Z}P$. Choose a compact real neighborhood for H around $[h_*]P$ on which the point coordinates, both difference charts, and the six tests are defined and bounded. Such a neighborhood exists: the rational functions restricted to the multiplication curve $H \mapsto (H, [n]H)$ are not identically singular. For every nonzero integer n, the multiplication map $[n]$ is nonconstant. The forbidden equalities $U = \pm A$ are equivalent to $h = \pm 1$, even when n is negative or even. Similarly, $H = \pm P$ means

$h = \pm 1$. Excluding $h = 0, \pm 1$ therefore handles both difference charts and all identity points. Every other nonzero multiple of the non-torsion P has nonzero abscissa and ordinate. Pass to a smaller compact neighborhood to avoid all remaining zero denominators. Enlarge L with $64 \mid L$ and the real component-group condition so that the progression $h \equiv h_* \pmod{L}$ preserves fixed lower valuation bounds for these tests at $2, 3, 17$. These are precisely the local-continuity and real-orbit steps of the source realization, with fewer point roles and no scalar constraint.

Choose a fixed D_0 with $D_0^2\theta/a_r \in \mathbb{Z}$ for all contacts and both scales, set $v = D_0^2 w^2$, and seek $\lambda = u/v$. Apply the retained prime-pattern theorem to the distinct forms $u - (\theta/a_r)v$, after repetitions have been removed. All bad primes and previously fixed moduli are excluded before this application. The theorem permits arbitrarily large divisibility of w, a sufficiently large prescribed lower bound on u/w^2, and the conjugate reduction-order conditions.

As in [3, 4], the annihilating order moduli U_0, V_0 for the two conjugate families satisfy

$$\gcd(U_0, V_0) = 1, \qquad \gcd(U_0, L) = 16, \qquad \gcd(V_0, L) = 1.$$

Thus the congruences $h \equiv 0 \pmod{U_0}$, $h \equiv 1 \pmod{V_0}$, $h \equiv h_* \pmod{L}$ are compatible, since $16 \mid h_*$. Choose h in that progression while retaining the real neighborhood and avoiding the finitely many excluded multiples. At one place above each contact prime, H, U are formal; at the conjugate place, $H - P, U - A$ are formal. Both exact logarithmic ratios are n. The constant approximation has precision four, so all six tests have their assigned order one, including the direct target $n = X$. The integer n is p-integral for every selected prime; oddness is not used. When $n = 1$ or -1, some comparison tests vanish identically, which is permitted: their factors in W_r equal one.

The inherited detector preparation now applies without alteration. For clarity, its main ingredients are the following. The identity

$$R_r = \frac{a_r^6 \prod_{\theta \in \Theta}(u - (\theta/a_r)v)}{v^6(a_r\lambda - 1)^8}$$

has a rational-square prefactor. In particular, the possibly nonsquare scale contributes the square a_r^6. At each chosen contact prime $v_p(R_r) = 1$; the successful conjugate place gives all inverse norm-factor residues equal to one, and the trace of W_r^{-1} is a unit with residue one or two. The resulting b_r has square reduced denominator, and the prescribed congruences give its local squareclasses at $2, 3, 17$. The fixed lower bounds on the tests and sufficient divisibility of w give the depths in (2). On the compact real set, uniformly,

$$R_r \sim a_r^{-2}\lambda^{-2}, \qquad W_r = 1 + O(\lambda^{-2}), \qquad b_r \sim (4a_r^2\lambda^2)^{-1}.$$

Thus $0 < b_r < 17$ for the sufficiently large lower bound on λ fixed before prime selection. All detector guards are nonzero. These calculations are the source comparison-pole preparation proof [4]; they depend on the finite list of tests and their orders, not on a Lucas or scalar test.

Each prepared input has an independent pole certificate with $pqs \neq 0$ by the retained pole-realization hypotheses. Since $p \neq 0$ excludes $e = 1$, the substitution $\tau = p/(e - 1)$ gives the single chart used above and its nonzero denominator. Thus both nonzero quadratic-root conditions are realized. The direct point charts cover A, H, U. Every nonzero integer input therefore has all required witnesses, for every fixed odd $B \geq 3$. The input zero is included by the polynomial operation in the next section.

7 The polynomial and its exact specification

The retained rational coordinates are

Coordinates	Number
Input X	1
Three point pairs for A, H, U	6
Comparison slope λ	1
Two independent pole pairs (τ_r, s_r)	4
Retained before finite-root elimination	12

There are three monic point quartics and two monic pole quadratics. The point algebra has rank $b = 4^3 = 64$, and the full algebra has rank $n_0 = 4^3 2^2 = 256$. Its guard has the form gq_1q_2, with g and both quadratic radicands in the point algebra. There are no further residual equations.

The separating-element theorem [7] replaces all five finite roots and every nonzero guard with one rational coordinate U_0. Let $Q_B^{(d)}$ be its exact integer polynomial. Define

$$F_B^{(d)} = XQ_B^{(d)}. \tag{9}$$

For $X \neq 0$, the rational zero sets of $F_B^{(d)}$ and $Q_B^{(d)}$ are the same. For $X = 0$, equation (9) is satisfied by every rational choice of the other coordinates. Soundness and completeness therefore give exactly $\mathbb{Z}$ for every sufficiently large fixed odd shift. The operation introduces no new coordinate, so the total remains thirteen.

For completeness, the finite formula for $Q_B^{(d)}$ is as follows. Let B_0 denote the point algebra, and A_0 the algebra after adjoining q_1, q_2, with $q_j^2 = c_j \in B_0$. Put

$$T_0 = g^2 c_1 c_2, \qquad \det_{B_0}(I - vM_{T_0}) = \sum_{j=0}^{64} a_j v^j, \qquad a_0 = 1.$$

Write $\theta_1, \ldots, \theta_5$ for the native generators and δ_i for the discriminant of the ith point quartic. The scalar polynomial

$$R_0 = 256 \left(1 + \sum_{i=1}^{5} \sum_{u,v=1}^{256} (M_{\theta_i}^{A_0})_{uv}^2 + \sum_{u,v=1}^{64} (M_g^{B_0})_{uv}^2 \right)$$

bounds all required eigenvalues. For each $1 \leq r \leq 64$, set

$$c_r = a_r, \qquad h_r(Z) = \sum_{j=0}^{r} a_j Z^{r-j}, \qquad E_r = c_r - h_r(T_0),$$

$$D_r = c_r^2 \prod_{i=1}^{3} \delta_i^2, \qquad s_r = 3r + 36,$$

$$K_r = (2R_0)^{s_r}(1 + c_r^2) \prod_{i=1}^{3} (1 + \delta_i^2), \qquad C_r = D_r + 4R_0 K_r,$$

$$\omega_r = D_r E_r \left(C_r^5 + \sum_{i=1}^{5} D_r^i C_r^{5-i} \theta_i \right), \qquad H_r(U_0) = \det_{A_0}(I - U_0 M_{\omega_r}).$$

Then

$$Q_B^{(d)} = H_{64}(U_0) \prod_{r=1}^{63} \left(H_r(U_0)^2 + \sum_{j=r+1}^{64} a_j^2 \right).$$

The guard idempotents isolate the nonzero part, and the squared tail coefficients select its correct degree. The separation theorem covers nonreal points, vanishing branches, and nilpotents on the inactive part. Thus this is a finite integer coefficient algorithm, not a prescription to select a favorable degree case.

7.1 Degree, support, and coefficient bounds

The independently checked native weights and guard weights are

d	generator weights	w_G	w_g
6	$(6, 6, 6, 51093, 65097)$	142557	26367
4	$(6, 6, 6, 51093, 60429)$	136845	25323.

The separating-element weights have the linear upper bound $W_1 r + W_0$, with

d	W_1	W_0
6	9146712	67163859
4	8718120	63704871.

Summing the principal-minor degree bounds over all 64 factors gives

$$\deg Q_B^{(d)} \le 256(W_1\, 64^2 + (W_0 + 1)127) =: \mathcal{D}_d.$$

The exact values are

$$\mathcal{D}_6 = 11774654098432, \qquad \mathcal{D}_4 = 11212784195584.$$

Equation (9) increases the degree by at most one, which proves the bounds in Theorem 1.1. The shift is a formal coefficient parameter of weight zero, and no branch or operation depends on its numerical value. The bounds therefore hold for every fixed odd $B \ge 3$.

The degree in U_0 is at most $E = 256 \cdot 127 = 32512$. Counting all monomials within these two degree bounds gives

$$\#\operatorname{supp}(Q_B^{(d)}) \le \binom{\mathcal{D}_d + 13}{13} - \binom{\mathcal{D}_d - E + 12}{13}.$$

Indeed monomials with U_0 exponent at least $E + 1$ are in bijection with monomials of degree at most $\mathcal{D}_d - E - 1$ after subtracting that exponent. Multiplication by X is an injective shift of exponent vectors, so it preserves the exact support size and the complete coefficient multiset. Exact integer calculation bounds the displayed expression by 2^{508} for $d = 6$, and by 2^{507} for $d = 4$, with strict inequalities. The determinant polynomials are not expanded, and no exact support count is asserted.

For the algebraic fixtures $B = 3$, the coefficient-norm certificates give

d	$\log_2 \|F_3^{(d)}\|_1$ upper bound
6	28598607557817
4	27756172773305.

Both numbers are below 2^{45}. These are fixture bounds only; neither $B = 3$ member is certified to define the integers.

7.2 Reconstruction and its scope

The module `computation/few_variables/direct_input13_v22` contains the source builder, the exact indexed specifications, and independent source and bound checkers. Its `weak` and `abc` directories contain the sixth- and fourth-power variants, respectively. Each source has 1756 rational-circuit nodes and 752 integer coefficient nodes. The exact conversion audit checks the three point quartics and their derivatives, two rational square clearings, and all 48 atomic guards. The changed scale and target identities are separately verified symbolically with B treated as a formal coefficient.

An independent implementation also reconstructs the complete rational formulas in fifteen finite-field samples per variant. These samples are bounded supplementary evidence. They do not replace the symbolic conversion, the general local proof, or completeness. The degree checker imports no producer, checks dependencies and companion-column inequalities, and sums all 64 degree cases independently. A final exact check verifies (9), its coordinate count, and its degree, support, and coefficient consequences. None of these computations verifies weak *abc* or the retained arithmetic premises.

8 The range of weak exponents

Corollary 8.1. *Retain the elliptic, pole, and prime-pattern inputs. Suppose weak rational abc holds for one exponent* $1 < \kappa < 4/3$. *Choose an even integer* d *such that*

$$\left(\frac{8}{\kappa} - 6\right)\left(\frac{d}{\kappa} - 3\right) > 1.$$

Direct comparison with X, *the scale* $(X^2+B)^d+T(n)^d$, *and the final factor* X *give a computable thirteen-variable family defining* $\mathbb{Z}$ *for all sufficiently large fixed odd shifts.*

Proof. Here $\gamma > 0$ and $\gamma\alpha > 1$. The height lemmas and local divisibility give $\eta \le 2(\gamma\alpha)^{-1}\log N + O(1)$. Lemma 2.3 makes the coefficient of η on the right equal to $(\gamma\alpha)^{-1} < 1$. The uniform soundness proof applies. Evenness of d preserves positivity, and completeness for nonzero integers is unchanged. Such an even d exists for every stated κ. □

For $d = 6$, the sufficient exponent range is

$$1 < \kappa < \frac{30 - 2\sqrt{21}}{17};$$

for $d = 4$, it is

$$1 < \kappa < \frac{24 - 4\sqrt{2}}{17}.$$

These follow by solving $\gamma\alpha > 1$ in the positive range. The first contains $121/100$ and the second contains $16/15$. The input-height separation does not extend the six-contact estimate beyond $\kappa < 4/3$: positivity of γ is still required. The numerical degree and support certificates above concern precisely the two supplied power scales.

References

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